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The Compression Theorem for Trained Neural Network Weights

Evidence-grounded FCN academic paper

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Abstract

We restate Shannon's source coding theorem in the form that applies to lossless compression of trained neural network weights, and combine it with the empirical value-level entropy of BF16-stored weights to produce a closed-form ceiling on any single-source lossless coder of those weights. For any lossless coder C applied to a sequence of N trained-weight values drawn from a source X with value-level entropy $H(X)$, the expected encoded size satisfies $\mathbb{E}|C(X)| \geq N \cdot H(X)$ bits. For BF16 trained weights with measured value-level entropy of approximately 10.51 bits per value, the corresponding compression-ratio ceiling for any single-source lossless coder is approximately $16/10.51 \approx 1.522\times$. We make explicit that this ceiling applies only to single-source coding without auxiliary con-

ditioning, that it does not constrain cross-model delta coders, content-addressable deduplication across artefact sets, or coders that change the reconstruction target, and that the numerical value of the ceiling depends on the measured $H(X)$ and on the storage format. The principal contribution of this paper is the regularity itself, applied to trained-weight data, together with the explicit scoping that distinguishes it from coding regimes that fall outside the theorem's hypotheses.

Keywords: lossless compression; neural network weights; Forlais compression research; Shannon source coding theorem; entropy; BF16

This is an approved public release of a Forlais Group academic paper.

Methodology Disclosure Boundary

This paper states a textbook information-theoretic result, restated for trained neural network weights, at the public-safe abstraction level. It reports measured quantities and scope. It does not disclose non-public Forlais implementation details, non-public code, run instructions, source logs, hidden benchmark mechanics, private datasets, prompts, model weights, infrastructure, or operational procedures.

1 Introduction

The most basic regularity in lossless compression is Shannon's source coding theorem [19], which fixes a lower bound on the expected code length of any uniquely decodable coder in terms of the entropy of the source. The theorem is a classical result with a textbook treatment [2] and is recovered in practice by any sufficiently strong entropy coder, including Huffman coding [6] and arithmetic coding [21, 17].

This paper restates the theorem in the specific form relevant to lossless compression of trained neural network weights. The motivation is internal: separate FCN entries report empirical compression ratios that need a common reference point against which they can be compared, and a precise statement of which regimes the theorem's lower bound actually constrains. We therefore record the theorem in its applicable form, combine it with the measured value-level entropy of BF16-stored trained weights, and state the resulting closed-form compression-ratio ceiling for single-source lossless coding of BF16 trained weights.

We make explicit that the resulting ceiling is a property of single-source coders, not of every coding regime relevant to neural network weights. Cross-model delta coders, content-addressable deduplication of artefact sets, and coders that change the reconstruction target each operate under different hypotheses, and the ceiling stated here is not a lower bound for any of them.

2 Scope

The result reported in this paper concerns:

- single-source lossless coding of a sequence of trained neural network weight values, with no auxiliary conditioning information;
- the application of Shannon's source coding theorem in this setting;
- the measured value-level entropy of BF16-stored trained weights from the FCN evidence base;
- the closed-form compression-ratio ceiling that follows from combining the theorem with the measured entropy.

The result does not concern:

- cross-model delta coders that condition on a related reference checkpoint, which are reported in separate FCN entries;
- content-addressable deduplication across sets of related artefacts, which is reported in separate FCN entries;
- lossy compression, pruning, distillation, or quantisation-aware retraining;

- universality across numeric formats; the closed-form ceiling depends on the storage format;
- rate-distortion regimes that change the reconstruction target [14].

3 Background and related work

Shannon’s source coding theorem [19] establishes that, for any discrete source X with entropy $H(X)$, every lossless prefix code C satisfies $\mathbb{E}|C(X)| \geq H(X)$ bits per source symbol, and the bound is attainable in the limit of long block codes. The standard textbook treatment, including the extension to uniquely decodable codes, is [2].

Practical entropy coders that approach the bound include Huffman coding [6], arithmetic coding [21, 17, 18], range coding [10], and asymmetric numeral systems [4]. Dictionary-based universal coders such as the Lempel-Ziv family [22, 23, 20] and general-purpose compressors such as DEFLATE, LZMA, Zstandard, and context-mixing variants [3, 16, 1, 9] combine dictionary matching with entropy coding.

Empirical entropy estimation on finite samples requires bias correction; the standard references are Miller’s correction [12] and Paninski’s treatment [15]. The IEEE 754 floating-point storage format relevant to BF16 trained weights is documented in [7, 8, 11].

Recent work on lossless compression of large language model weights includes dynamic-length floating-point methods [5], which target the same per-value distribution that the theorem here bounds.

4 The compression theorem, restated

Let $X_1, X_2, \dots, X_N$ be the value sequence of a trained neural network weight tensor stored in a fixed numeric format of w bits per value, so that the raw byte budget of the tensor is $w \cdot N$ bits. Assume the values are drawn from a source distribution with value-level entropy $H(X)$, measured in bits per value, where the entropy is taken with respect to the same per-value alphabet as the storage format.

Statement (single-source compression theorem). For any uniquely decodable lossless code C applied to the sequence $X_1, \dots, X_N$, the expected encoded length satisfies

$$\mathbb{E}|C(X_1, \dots, X_N)| \geq N \cdot H(X) \text{ bits.} \quad (1)$$

The expected compression ratio relative to the stored byte budget therefore satisfies

$$\mathbb{E} r = \frac{w \cdot N}{\mathbb{E}|C(X_1, \dots, X_N)|} \leq \frac{w}{H(X)}. \quad (2)$$

The bound in Equation (1) is the textbook statement of Shannon’s source coding theorem [19, 2] applied to a sequence of independent or stationary ergodic per-value samples. The reformulation in Equation (2) simply expresses the same bound as a ceiling on the per-source compression ratio rather than as a floor on the per-source encoded length.

Hypotheses. Equation (1) requires three hypotheses: the coder is uniquely decodable, the coder operates only on the value sequence with no auxiliary conditioning, and the encoded representation reconstructs the original value sequence bit-for-bit. The first hypothesis is standard. The second hypothesis is the one that distinguishes the regime covered by this paper from cross-model delta coders, deduplicating archives, and rate-distortion regimes, and we restate it explicitly in the scope section above. The third hypothesis is enforced in the FCN verification protocol by a SHA-256 round-trip [13].

Single-source ceiling for BF16 trained weights. For the BF16 storage format, $w = 16$ bits per value. The FCN evidence base reports a measured value-level entropy of approximately $H(X) \approx 10.51$ bits per value for trained weights in the present evidence set. Substituting into Equation (2) gives

$$r_{\max}^{\text{BF16}} \leq \frac{16}{10.51} \approx 1.522 \times . \quad (3)$$

Equation (3) is the closed-form ceiling for any single-source lossless coder of BF16 trained weights in the tested evidence set. The numerical value $1.522 \times$ depends on the measured $H(X)$ and on the storage format; the structural form $w/H(X)$ depends only on the theorem and the storage width.

5 Verification protocol

The theorem in Equation (1) is a mathematical statement and does not require empirical verification in this paper. The substitution that produces the closed-form ceiling in Equation (3), however, depends on the measured value-level entropy $H(X) \approx 10.51$ bits per value, which is itself produced by a verification protocol described in a separate FCN entry. We summarise the protocol for completeness:

- per-value samples are drawn from full-tensor BF16 trained weight artefacts in their stored format;
- the per-value empirical distribution is constructed at the value-level alphabet of BF16;
- a bias-aware entropy estimator is applied per tensor [12, 15];
- the per-tensor estimates are aggregated across all tensors and across multiple architectures in the evidence set;
- the aggregated value of approximately 10.51 bits per value is reported with the cross-tensor standard deviation in the separate FCN entry.

This paper takes the aggregated $H(X) \approx 10.51$ bits per value as an input and reports only the closed-form consequence under Equation (2).

6 Empirical reconciliation

Empirical compression ratios reported in separate single-source FCN entries on BF16 trained weights cluster below the closed-form ceiling in Equation (3). The strongest single-source ratios fall in the immediate neighbourhood of $1.522 \times$ without exceeding it, which is exactly the behaviour the theorem predicts under its hypotheses.

Empirical compression ratios reported in separate FCN entries for cross-model delta coders and for content-addressable deduplication exceed $1.522 \times$, sometimes by a substantial margin. This is not a contradiction of Equation (3): those entries report coders that condition on additional information or operate over different units of measurement, and they fall outside the second hypothesis of the theorem.

7 Theoretical analysis: a refined statement and its tightness

We record two refinements of the statement in Equation (1) that bear on its interpretation in the FCN setting.

Block-coding tightness. Shannon’s theorem is tight only in the limit of long block codes. For finite N , the gap between $\mathbb{E}|C(X_1, \dots, X_N)|$ and $N \cdot H(X)$ is bounded above by an additive constant that depends on the coder family. For arithmetic coding over a known distribution, the overhead is at most two bits per coding interval [21, 17], which is negligible at the per-tensor scale relevant to neural network weights. For Huffman coding, the per-symbol overhead is at most one bit, with the worst case attained only for highly skewed distributions [6]. The closed-form ceiling in Equation (3) is therefore tight in the practical sense for any sufficiently strong entropy coder operating on the value-level alphabet.

Stationarity and independence assumptions. The expectation in Equation (1) is taken over the source distribution, and the per-symbol entropy $H(X)$ is defined for a stationary source. Trained neural network weight sequences are not strictly stationary: per-tensor distributions vary materially with the tensor role, and a single tensor is itself a finite realisation rather than a sample from an infinite stationary process. The aggregate $H(X) \approx 10.51$ bits per value reported in the separate FCN entry is a cross-tensor empirical mean. Per-tensor entropies vary by role, but the cross-tensor variation is small enough that the aggregate ceiling in Equation (3) applies to all tested tensor roles within measurement-level standard deviation. We do not claim that the same ceiling applies to architectures we have not measured.

Universal coders. Universal lossless coders such as the Lempel-Ziv family [22, 23] attain the entropy rate of any stationary ergodic source in the limit, without knowledge of the source distribution. This is the strongest statement of Shannon-style tightness that applies to general-purpose compressors and is the reason high-quality general-purpose coders such as [3, 1, 16] approach but do not exceed the entropy bound on neural network weight tensors. The fact that the strongest single-source coders in the FCN evidence base cluster just below $1.522\times$ without exceeding it is consistent with this universality.

8 Connection to public literature

The closed-form ceiling for BF16 single-source coding is consistent with the practical behaviour of mature entropy coders on trained weights documented in independent published work, including dynamic-length floating-point methods [5]. The independent published ratios for single-source coders of trained BF16 weights also approach but do not exceed the same neighbourhood. We report this as a consistency check rather than as an additional measurement.

9 Implications

The single-source ceiling has three consequences for the FCN evidence base.

First, any future FCN entry that reports a single-source coder of BF16 trained weights with a ratio materially above $1.522\times$ must either revise the measured $H(X)$ used in Equation (3), identify auxiliary conditioning that places the coder outside the single-source regime, or identify a verification failure in the coder. A nominal ratio above the ceiling without one of these three justifications cannot stand.

Second, regimes that materially exceed $1.522\times$ in independent FCN entries, including cross-model delta coders and content-addressable deduplication of artefact sets, are not in tension with the theorem; they are in different regimes, and the headline ratios from those regimes should not be combined with single-source ratios without explicit re-scoping.

Third, the closed-form structure $w/H(X)$ in Equation (2) indicates the two levers that change the ceiling: the storage format width w and the value-level entropy $H(X)$. Format redesigns and rate-distortion regimes both change one of the two levers, but they do so by changing the reconstruction target, and are outside the scope of this paper.

10 Cross-format observations

The structural form $r_{\max} \leq w/H(X)$ applies to any IEEE 754 storage format of w bits per value, with $H(X)$ recomputed for the value distribution under that format. For FP32 storage, with $w = 32$ bits per value, the measured per-value entropy of trained weights in the same evidence set is larger than for BF16 because the additional mantissa bits expose additional fine-scale variation, and the resulting closed-form ceiling differs from the BF16 ceiling. We do not report the FP32 numerical ceiling here; it is recorded in a separate FCN entry. The point of mentioning the cross-format observation is to make clear that the closed-form ceiling does not have a single number that applies across formats: changing the storage width changes both the numerator and the value-level entropy in the denominator, and the two effects are not independent.

11 Discussion

The compression theorem is a textbook regularity, and this paper makes no claim of novelty in the theorem itself. The contribution is the explicit restatement in the form that applies to BF16 trained weights and the explicit scoping that distinguishes the single-source regime from cross-model and cross-artefact regimes elsewhere in the FCN evidence base. The closed-form ceiling $r_{\max}^{\text{BF16}} \approx 1.522 \times$ is reported as the natural consequence of combining the theorem with the measured $H(X) \approx 10.51$ bits per value from a separate FCN entry. The theorem itself is unchanged from [19]; the application to a specific measured source is what makes the ceiling concrete.

12 Empirical reconciliation across the FCN evidence base

We summarise three reconciliations between the closed-form ceiling in Equation (3) and the empirical FCN evidence base.

Single-source coders on BF16 trained weights. The strongest empirical ratios reported in single-source FCN entries on BF16 trained weights sit immediately below $1.522 \times$. The closed-form ceiling is therefore tight in the practical sense: a sufficiently strong entropy coder operating on the value-level alphabet, with no auxiliary conditioning, attains a ratio that matches the bound to within the cross-tensor measurement-level standard deviation of $H(X)$. This is the textbook tightness of Shannon's theorem [19, 2] reflected in the FCN measurements.

Cross-model delta coders. The strongest empirical ratios reported in cross-model delta FCN entries exceed $1.522 \times$ on paired BF16 checkpoints. These coders condition on a related reference checkpoint and operate over a conditional distribution whose entropy is smaller than the unconditional value-level entropy $H(X)$. The closed-form ceiling in Equation (3) does not apply, because the second hypothesis of the theorem (no auxiliary conditioning) is violated by construction.

Deduplication of related artefact sets. The empirical aggregate factors reported in deduplication FCN entries exceed $1.522\times$ by a substantial margin. These factors are not per-source compression ratios: they are aggregate factors over an artefact set, with the unit of measurement being the set rather than any single source. The closed-form ceiling does not apply because the regime is outside the per-source coding setting that the theorem is about.

The three reconciliations together confirm that the theorem and its closed-form consequence are doing the work expected of them: they constrain the coders that fall under their hypotheses and remain silent about coders that do not.

13 Limitations

We note five limitations.

1. The closed-form ceiling depends on the measured $H(X) \approx 10.51$ bits per value reported in a separate FCN entry. Any revision to that measurement shifts the numerical value of the ceiling.
2. The theorem's lower bound assumes a uniquely decodable coder operating on the value sequence with no auxiliary conditioning. Coders that fall outside these hypotheses are not constrained by this paper.
3. The numerical ceiling is specific to BF16 storage. The structural form $w/H(X)$ applies to any format of w bits per value, but $H(X)$ is a property of the value distribution under that format and is not interchangeable across formats.
4. The bound is an expected-value statement. Single tensors may attain ratios slightly above or below the expected ceiling in finite samples without contradicting the theorem.
5. Public reproducibility of the underlying $H(X)$ measurement is limited by disclosure control on the weight artefacts. The protocol can be reproduced in principle on any IEEE 754 trained-weight artefact under the same bias-aware estimator.

14 Data and code availability

Per-experiment records, verification artefacts, and raw measurements are maintained by Forlais Group. Materials can be made available on reasonable request subject to disclosure review, confidentiality restrictions, and removal of proprietary methodology or operational detail. This paper does not promise public release of any internal artefact beyond the abstraction level reported here.

15 Competing interests

The author is affiliated with Forlais Group Ltd, which conducts the FCN compression research programme described in this paper. The author declares no other competing financial or non-financial interests.

16 Conclusion

The FCN evidence base treats Shannon's source coding theorem [19, 2] as the fundamental regularity that bounds single-source lossless coding of trained neural network weights from below. Combined

with the measured value-level entropy of BF16 trained weights of approximately 10.51 bits per value, the theorem produces a closed-form compression-ratio ceiling of approximately $1.522\times$ for any single-source lossless coder of BF16 trained weights, as expressed in Equation (3). The ceiling applies to single-source coders without auxiliary conditioning and is not a lower bound for cross-model delta coders, content-addressable deduplication, or rate-distortion regimes, each of which is reported in separate FCN entries.

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Revision History

Version	Date	Change summary	Owner
0.1	14 May 2026	Initial FCN paper package generated from source evidence.	Oliver Cooper
0.2	14 May 2026	Added closed-form ceiling and hypothesis restatement.	Oliver Cooper
1.0	14 May 2026	Full evidence-grounded manuscript with theorem restatement, BF16 substitution, and regime scoping.	Oliver Cooper